

Study suggests that sexual behavior of infant capuchin monkeys may have complex functions

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## Actions displayed during infancy may be associated with maturation, the establishment of affective bonds and dominance

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The capuchin monkeys were observed in their natural environment, and the sexual behavior they displayed during the first year of life may not be associated with practice for reproductive roles in adulthood – Photo: Laboratory of Ethology, Development and Social Interaction/IPUSP

A study that analyzed 265 hours of recordings of wild infant capuchin monkeys suggests that the sociosexual behaviors they display as early as their first year of life perform functions that go beyond reproduction. The results contradict the hypothesis that these habits would only serve as a test for reproductive roles in adulthood. According to the scientists, the practices may be associated with the development of sex-specific characteristics, such as hormonal maturation and broader social functions, including affective bonds, the establishment of dominance hierarchies and the refinement of motor skills.

The study involving 16 infants of the species *Sapajus libidinosus* and *Sapajus xanthosternus* was conducted by scientists from USP's Institute of Psychology (IP) and Institute of Biosciences (IB), in collaboration with researchers from the University of Lethbridge, Canada, and the National Institute of Advanced Studies, India. The results were [published](<https://www.google.com/search?q=https://www.sciencedirect.com/science/article/pii/S1090513826000139>) in an article in the journal *Evolution and Human Behavior*.

## Males in infancy, females in adulthood – a reversal of leading roles

Male infants initiate sexual behaviors earlier than females and display these behaviors with greater frequency and variety, according to the study. Psychologist and anthropologist Irene Delval, from IP, explained that “the finding contrasts with the pattern observed in adult primates, when courtship is led by females.”

According to the author of the study, some [studies](<https://www.scopus.com/pages/publications/0036223380>) have already demonstrated this performance: female capuchin monkeys, during the mating season, actively solicit the presence of males – generally the alpha male 0 through facial expressions, body postures, vocalizations and movements.

For biologist Solimary Garcia Hernández, from IB, responsible for the statistical analysis of the study data, the most surprising result was the reversal of the pattern observed in

adult capuchin monkeys.

Irene Delval - Photo: Courtesy of Irene Delval

While adult females lead courtship interactions, among infants, males displayed a higher frequency of sociosexual behaviors. The researcher emphasizes that the finding challenges the idea that these behaviors in infancy are merely preparation for adulthood.

The researchers evaluated 23 sexual behaviors present in the heterosexual repertoire of adult primates. Among them, six appeared more frequently and were examined in detail: erection, mounting, mounting attempt, eyebrow raising, genital inspection and chest rubbing. The contexts in which these behaviors occurred most frequently and the partners preferred by infants during interactions were also investigated.

### ## Sexual behaviors

Erection was the most frequent sexual behavior in the two species analyzed, especially among males, occurring more often than other observed practices, such as mounting, mounting attempt, eyebrow raising, genital inspection and chest rubbing. According to Irene Delval, in females it was not possible to observe clitoral erection directly, because the clitoris is less exposed than the penis. It was recorded only indirectly during interactions in which females acted as recipients of male erections.

Capuchin monkeys observed in the forest – Photo: Laboratory of Ethology, Development and Social Interaction/IPUSP

Mounting was common in both sexes, being more frequent among males. Only “mounting attempt” was more relevant among females. As for the other behaviors, including mounting attempt, genital inspection, chest rubbing and eyebrow raising, all were observed in males and females, but were more frequent in males. Masturbation, genital display and body washing with one’s own urine appeared almost exclusively among males. Affiliative behaviors, such as hugging, grooming and outstretched arms, showed no relevant differences between the sexes.

The study concluded that age and sex influence the development of sexual behaviors in distinct ways, varying according to the type of behavior analyzed. The results showed that males display a higher frequency of sexual behaviors than females, and this difference tends to increase further with age. In addition, the data indicate that males are predominantly the emitters of these behaviors, while females appear more frequently as recipients.

Male infants interacted mainly with other males, with adults as their most frequent partners, followed by juveniles and, lastly, other infants. Females, in turn, interacted more with male infants, as well as juvenile females and, less frequently, other infants.

According to Irene Delval, the results suggest that sexual behaviors in infancy perform distinct social functions among males and females. In males, these interactions appear to be related to the formation of social bonds and the regulation of hormonal arousal. Among females, the behaviors occurred more in contexts of exploration and play, without such a clearly defined social function.

The study also analyzed the contexts in which sexual interactions among infants occurred and identified play as the main setting for these practices, in both sexes, with greater frequency among males. Explicitly sexual interactions came next, followed by activities such

as foraging and nursing. Although less common, these behaviors were also recorded in social contexts, such as mutual care and situations of conflict and threat.

Solimary Garcia Hernández - Photo: Courtesy of Solimary Garcia Hernández

### ## Parallels with human sexuality

According to the researchers, understanding how behavioral sexuality emerges in the first months of primates' lives may help explain the flexibility and development of sexuality in different species, including humans. Irene emphasizes that children also manifest curiosities, feelings and play related to sexuality, although these practices are often inhibited by cultural and religious norms, unlike what occurs among animals.

For Solimary Hernández, "observing how capuchin monkeys express sexual behaviors so naturally from the first weeks of life helps us demystify our own sexuality."

The article \*[Early sex differences in sociosexual behavior of wild robust capuchin monkeys: Ontogenetic and evolutionary insights](<https://www.google.com/search?q=https://www.sciencedirect.com/science/article/pii/S1090513826000139>)\* was published in the journal \*Evolution and Human Behavior.\*

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